IOT 2025/2026 - Challenge 2

Part 2: exercise

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Data provided)

COAP_CON_PUT_SIZE = 70 B = 560 b

COAP_PUT_ACK_SIZE = 15 B = 120 b

MQTT_SUB_SIZE = 65 B = 520 b

MQTT_SUB_ACK_SIZE = 55 B = 440 b

MQTT_PUB_SIZE = 75 B = 600 b

MQTT_PUB_ACK_SIZE = 50 B = 400 b

MQTT_CONNECT_SIZE = MQTT_DISCONNECT_SIZE = 60 B = 480 b

MQTT_CONNECT_ACK_SIZE = 50 B = 400 b

E_TX = 50 nJ/b

E_RX = 58 nJ/b

EQ1)

MEASUREMENTS_PER_HOUR = 60 min / 2 min = 30

a) COAP (direct communication)

E_COAP =
= MEASUREMENTS_PER_HOUR * (E_TX + E_RX) * (COAP_CON_PUT_SIZE + COAP_PUT_ACK_SIZE) =
= 30 * (50 nJ/b + 58 nJ/b) * (560 b + 120 b)
= 2203200 nJ = 2203,2 µJ

b) MQTT (via the gateway)

If we ignore the energy consumption of the gateway, as it is requested, we get that the total energy consumed by the sensor and the actuator is the sum of the energy spent for the initial connection of both the devices to the gateway and the energy spent for all the cycles completed in one hour.
For each cycle, in particular, these are the contributes:

- the energy used by the sensor to publish the message and receive the relative acknowledgement,
- the energy used by the actuator to receive the message published by the broker and send the acknowledgement.

E_MQTT =
= 2 * (MQTT_CONNECT_SIZE * E_TX + MQTT_CONNECT_ACK_SIZE * E_RX) +
(MQTT_SUB_SIZE * E_TX + MQTT_SUB_ACK_SIZE * E_RX) +
MEASUREMENTS_PER_HOUR *
[(MQTT_PUB_SIZE * E_TX + MQTT_PUB_ACK_SIZE * E_RX) +
(MQTT_PUB_SIZE * E_RX + MQTT_PUB_ACK_SIZE * E_TX)] =
= 2 * (480 b * 50 nJ/b + 400 b * 58 nJ/b) +
(520 b * 50 nJ/b + 440 b * 58 nJ/b) + 30 *
[(600 b * 50 nJ/b + 400 b * 58 nJ/b) +
(600 b * 58 nJ/b + 400 b * 50 nJ/b)] =
= 3385920 nJ = 3385,92 µJ

EQ2)

MQTT with QoS=1 and persistent conn. for actuator: no retransmissions so lower energy consumed. Sensor, once connected to broker, publishes (*retain=false*) every 2min; broker buffers samples; actuator connects to broker (*CleanSession=false*), receives samples (HP: it needs all of them) and disconnect.